MCU

文件: MCU.kicad_sch

POWER_SHARED

文件: power.kicad_sch

FOC_M1

文件: FOC_M1.kicad_sch

FOC_M2

文件: FOC_M2.kicad_sch

Dual independent motor framework
Common input supply + GD32F505VxTx controller
Review protection thresholds and ratings before energizing.

HSL_DEBUGGER

文件: HSLink.kicad_sch

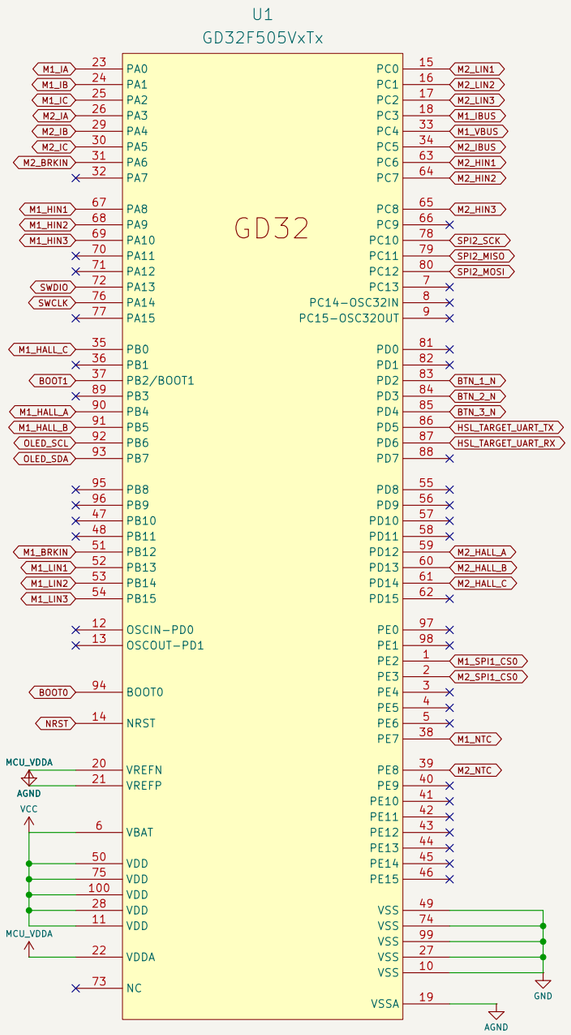
INTERFACE

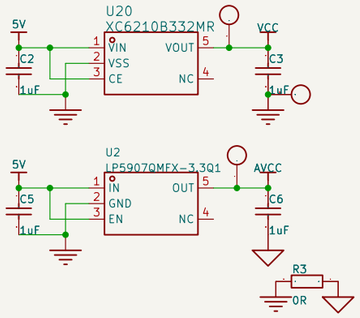
文件: Interface.kicad_sch

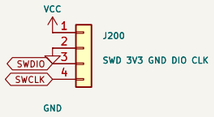
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File: GD32F5_FOC.kicad_sch

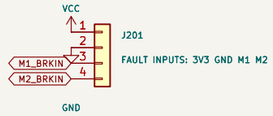
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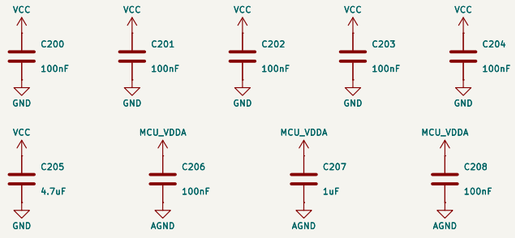
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 1/7



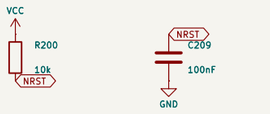


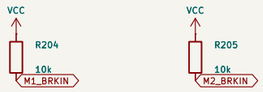












Onboard debugger: HSLink page; R8901..R8905 are default-open.
J200 external probe: never use simultaneously with onboard probe.
USART1 bridge PD5/TX, PD6/RX; target must be powered before enabling links.

MCU_VDDA = VCC through R203 (0R), bypass C206..C208 to AGND.
AVCC LDO remains for external analog amplifiers only.
J201: 1=3V3, 2=GND, 3=M1_BRKIN, 4=M2_BRKIN.
FAULT inputs use pullups; external open-drain HEALTH holds LOW only while healthy.
Configure each timer break polarity HIGH; unplugged defaults to brake. No autonomous trip comparator fitted.
Current limits/thresholds and external fault source remain to be designed.

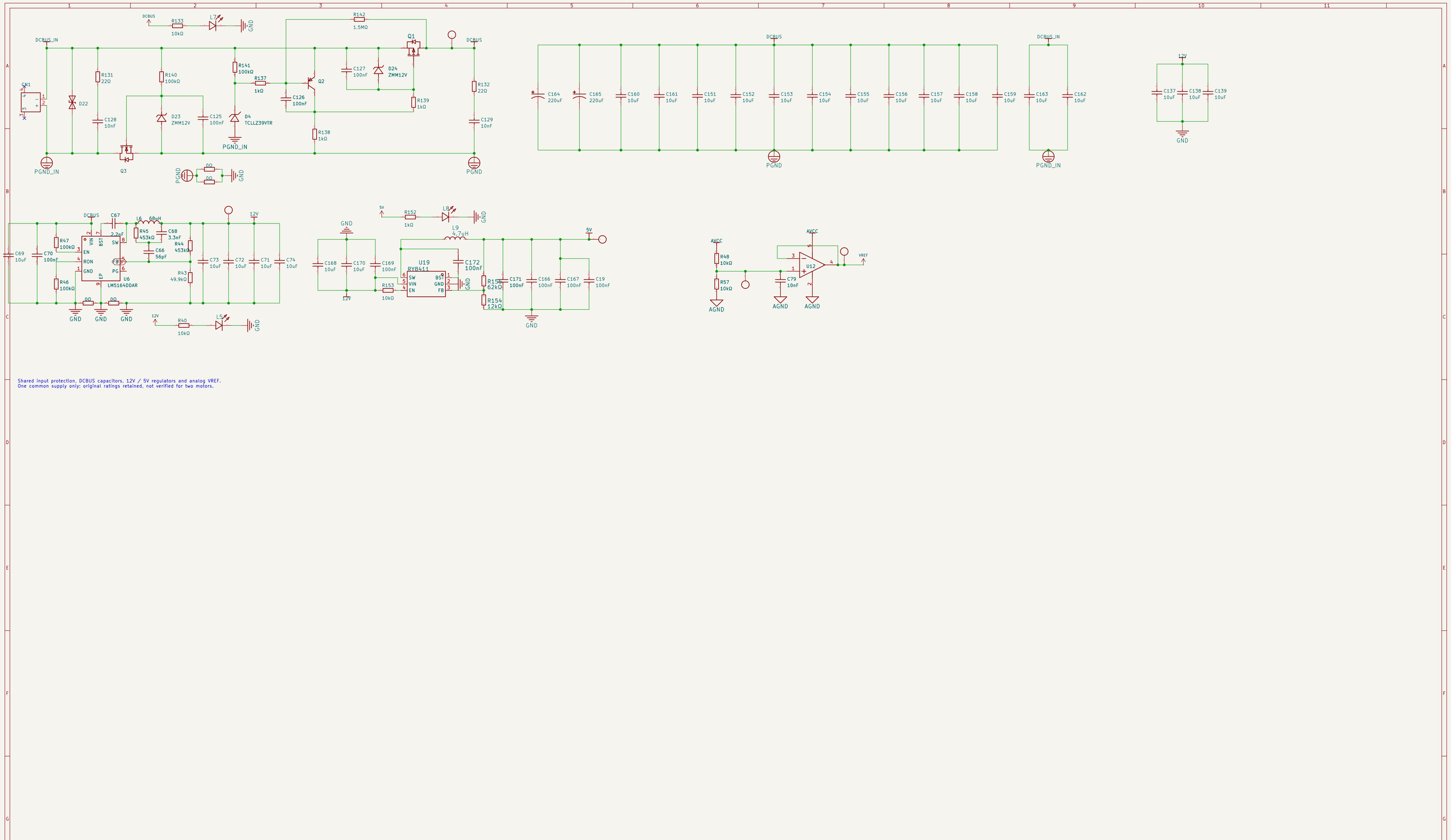
OLED1.29: I2C0 PB6 AF1 SCL / PB7 AF1 SDA.

Reference RC reset; PA7/PB3/PE4/PE5 unused (NC).

Buttons PD2/PD3/PD4 active-low. See Interface page.

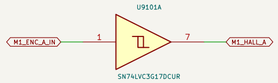
M1 TIMER0: PA8/9/10 AF1; PB13/14 AF2, PB15 AF1. BRKIN PB12 AF4.
M2 TIMER7: PC6/7 AF3, PC8 AF2; PC0/1/2 AF0. BRKIN PA6 AF4.
M1 currents PA0/1/2; M2 currents PA3/4/5 (ADC0/ADC1).
IBUS: PC3 / PC5; VBUS PC4; NTC PE7 / PE8 (ADC2).
Hall M1 PB4/5/0 TIMER2; M2 PD12/13/14 TIMER3. Release PB4 JTAG, keep SWD.
Shared SPI2 PC10/11/12 AF2; GPIO chip selects PE2 / PE3.
VA/VB/VC/COM and M2_VBUS stay on power-sheet testpoints, not MCU ADC.

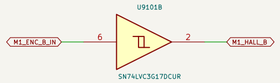
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Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 2/7

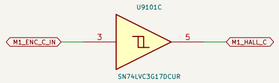


Shared input protection, DCBUS capacitors, 12V / 5V regulators and analog VREF.
One common supply only; original ratings retained, not verified for two motors.

Sheet: /POWER_SHARED/	
File: power.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4	**Rev:** Id: 3/7

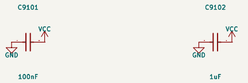








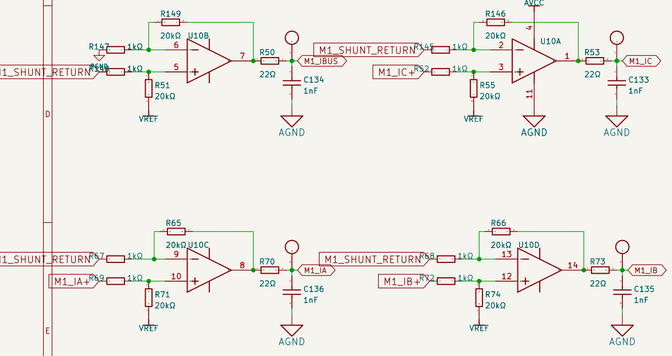
CN5: Hall / single-ended ABZ (one mode at a time)
1 GND; 2 C / Z; 3 B; 4 A; 5 auxiliary PWM (unused); 6 VSUP; 7/8 shield GND.
Default: 3.3V power and 3.3V push-pull signals. VCC = 3.3V.
10k pullups, 1k pulldowns and 100pF EMI capacitors DNP.
Fit pullups only for open-drain after cable / edge-rate review.
Pin 6 supply: fit ONE OR selector only; 5V option DNP.
5V Hall / encoder operation pending TI datasheet validation.
Buffer pinout from KiCad library; silicon / footprint review required before fabrication.

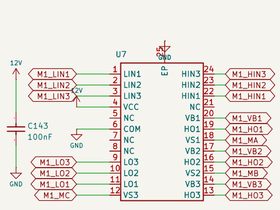


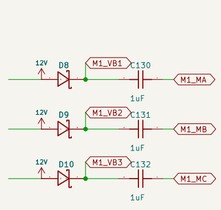


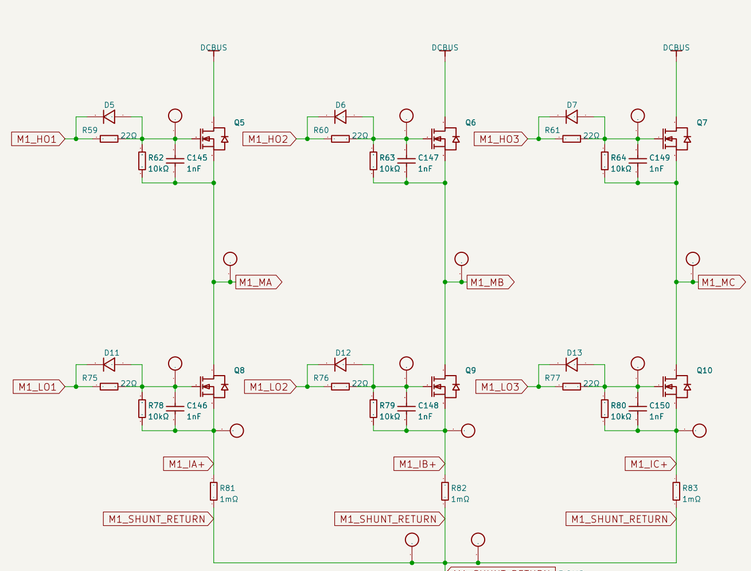


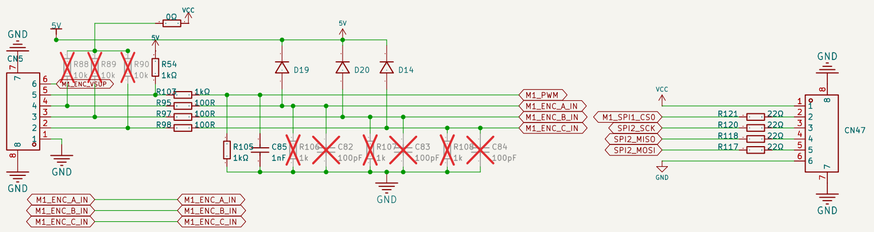
Buffer decoupling: 100nF + 1uF close to pins 8 / 4.
Existing 5V input clamps retained; power sequencing not validated.

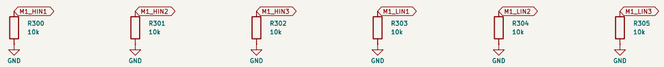








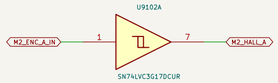


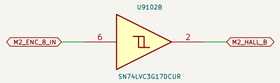


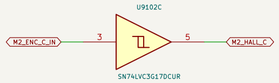
MOTOR 1: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M1_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

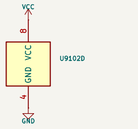
VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

Sheet: /FOC_M1/
File: FOC_M1.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 4/7









CN1005: Hall / single-ended ABZ (one mode at a time)
1 GND; 2 C / Z; 3 B; 4 A; 5 auxiliary PWM (unused); 6 VSUP; 7/8 shield GND.
Default: 3.3V power and 3.3V push-pull signals. VCC = 3.3V.
10k pullups, 1k pulldowns and 100pF EMI capacitors DNP.
Fit pullups only for open-drain after cable / edge-rate review.
Pin 6 supply: fit ONE 0R selector only; 5V option DNP.
5V Hall / encoder operation pending TI datasheet validation.
Buffer pinout from KiCad library; silicon / footprint review required before fabrication.

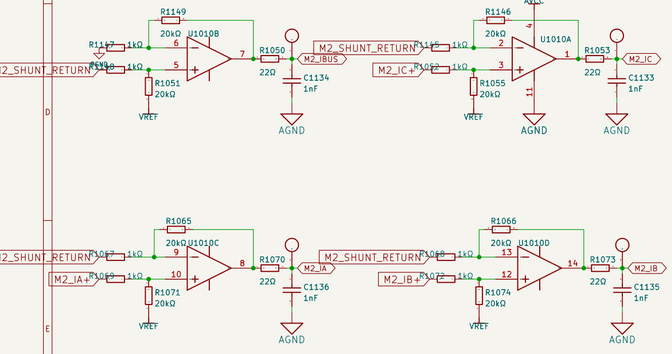


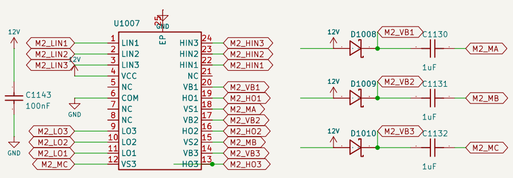


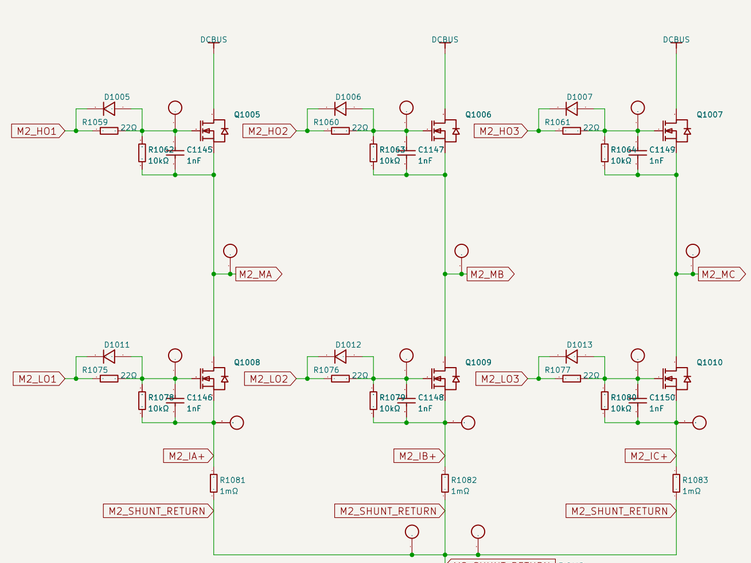


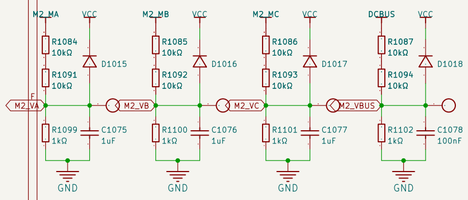


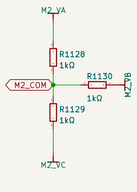
Buffer decoupling: 100nF + 1uF close to pins 8 / 4.
Existing 5V input clamps retained; power sequencing not validated.

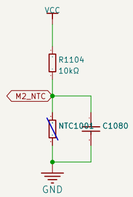


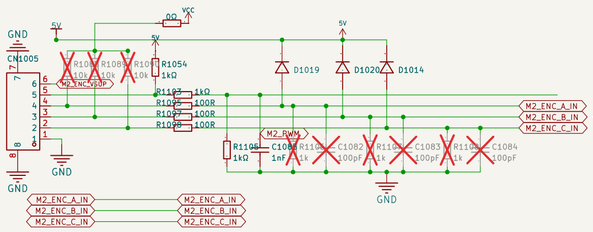


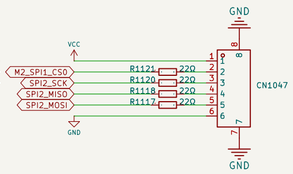


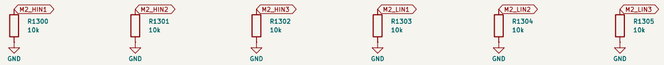










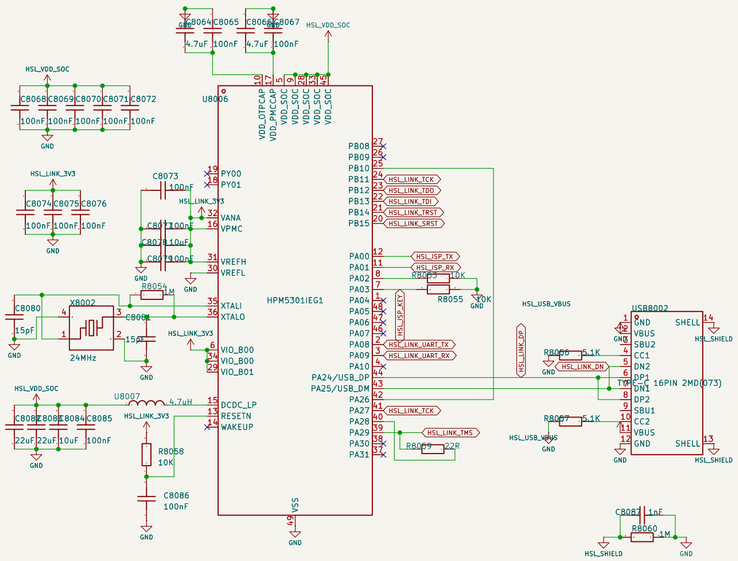


MOTOR 2: independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M2_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

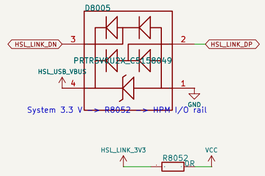
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File: FOC_M2.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 5/7

HSLINK / HPM5301 — SHARED SYSTEM POWER

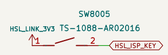


HPM5301 + 24 MHz clock + on-chip VDD_SOC buck

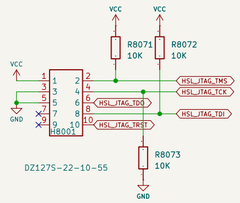
USB data / ESD: VBUS does not power board



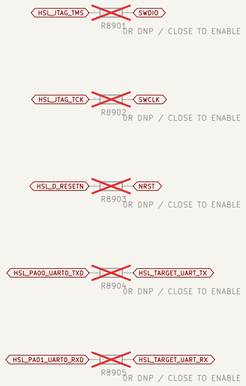
ISP boot button



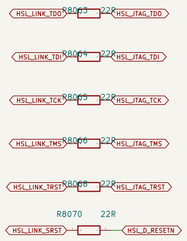
Target reference header



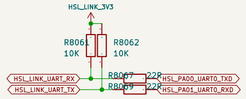
Target disconnect links (default DNP)



JTAG series termination



UART series termination



POWER: VCC = existing system 3.3 V (U20 XC6210B332MR).
R8052 OR supplies HSL_LINK_3V3: HPM on-chip VDD_SOC buck retained.
USB VBUS only reaches USB connector / ESD; no board-supply path.
U20 load / thermal margin requires measurement; HPM load is unknown.

R8901..R8905 remain DEFAULT DNP / OPEN. Fit only with target powered.
Never drive target simultaneously from onboard debugger and J200 probe.
HB001 pin 1 is target VCC reference; no separate power injection.
No validated ADC voltage sensing or firmware power-off tri-state.

Reference GPIO mapping retained: firmware compatibility NOT verified.
USB footprint and firmware require validation before PCB fabrication.

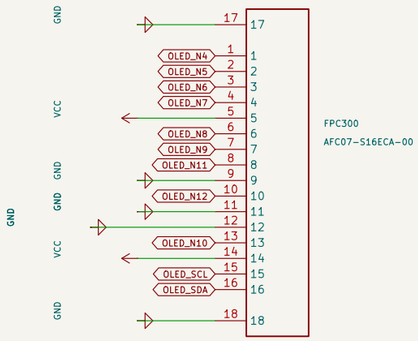
UART nets name target direction: TX = GD32 PD5 -> probe RX;
RX = GD32 PD6 <- probe TX.

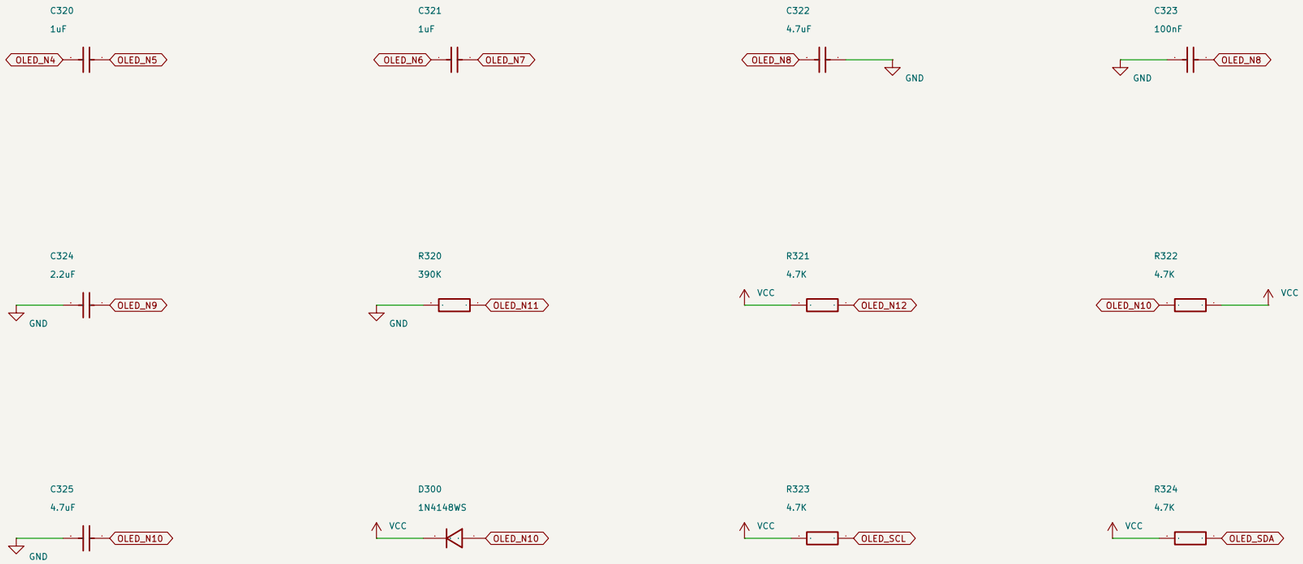
Sheet: /HSL_DEBUGGER/
File: HSLink.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 6/7

OLED1.29 REFERENCE / 16-PIN 0.5mm FPC / 3.3V

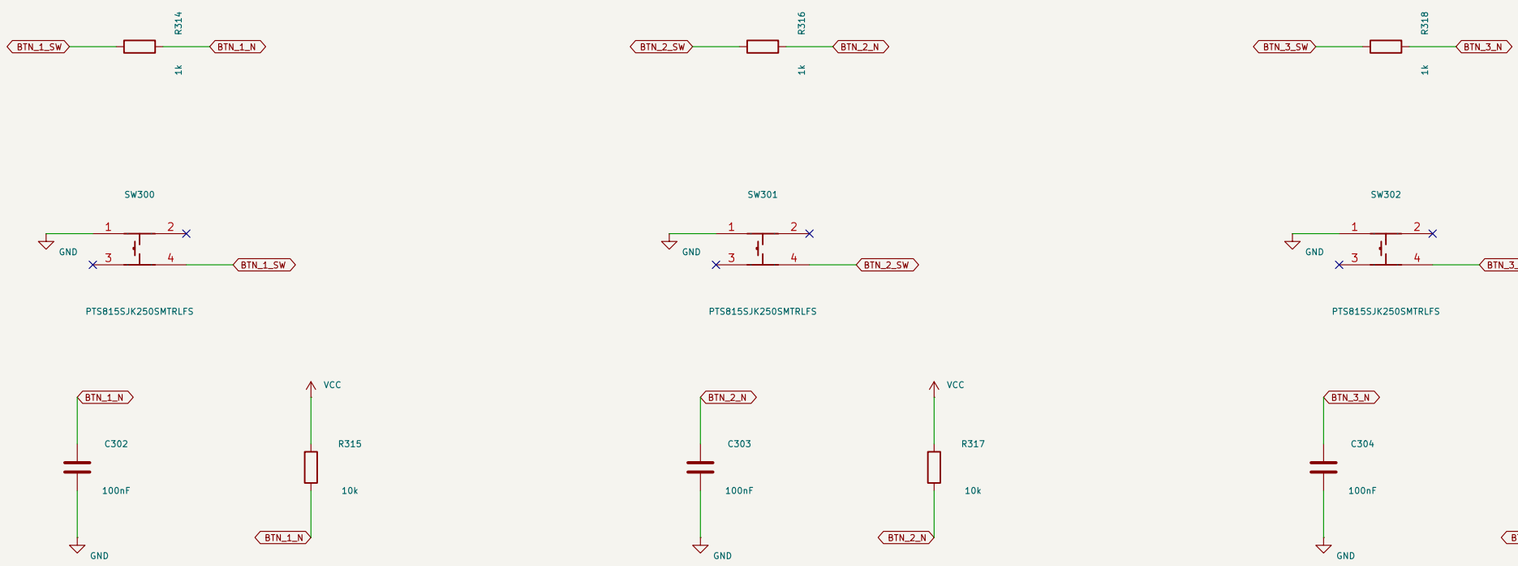
FPC300: AFC07-S16ECA-00; mounting pads 17/18 grounded.

I2C0: PB6 AF1 = SCL; PB7 AF1 = SDA. Original hardware straps + RC reset retained.





THREE ACTIVE-LOW LOCAL BUTTONS: PD2 / PD3 / PD4



SW300..302 reuse reference SW5/SW6 C&K symbol and footprint; pins 2/3 left NC as reference.

10k pull-up + 100nF at GPIO; 1k switch-series resistor limits capacitor discharge.

Pressed level approximately 0.30V with 3.3V supply. Apply firmware debounce. No external button cable.

Sheet: /INTERFACE/		
File: Interface.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 7/7